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PROSER: A Web-Based Peripheral Blood Smear Interpretation Support Tool Utilizing Electronic Health Record Data

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ABSTRACT

Objectives: Peripheral blood smear (PBS) interpretation represents a cornerstone of pathology practice and resident training but has remained largely static for decades. Here, we describe a novel PBS interpretation support tool.

Methods: In a mixed-methods quality improvement study, a web-based clinical decision support (CDS) tool to assist pathologists in PBS interpretation, PROSER, was deployed in an academic hospital over a 2-month period in 2022. PROSER interfaced with the hospital system's electronic health record and data warehouse to obtain and display relevant demographic, laboratory, and medication information for patients with pending PBS consults. PROSER used these data along with morphologic findings entered by the pathologist to draft a PBS interpretation using rule-based logic. We evaluated users' perceptions of PROSER with a Likert-type survey.

Results: PROSER displayed 46 laboratory values with corresponding reference ranges and abnormal flags, allowed for entry of 14 microscopy findings, and computed 2 calculations based on laboratory values; it composed automated PBS reports using a library of 92 prewritten phrases. Overall, PROSER was well received by residents.

Conclusions: In this quality improvement study, we successfully deployed a web-based CDS tool for PBS interpretation. Future work is needed to quantitatively evaluate this intervention's effects on clinical outcomes and resident training.

INTRODUCTION

Peripheral blood smear (PBS) interpretation represents a fundamental skill learned in pathology resident physician training,[1] practiced by pathologists and hematopathologists worldwide. It is an essential component of the workup of anemia, pancytopenia, neutrophilia, and other laboratory abnormalities and suspected medical conditions.[2-4] Despite its importance, PBS interpretation has remained largely unchanged for decades and continues to be a time- and resource-intensive procedure,[5] often involving significant clerical burden on the part of the physician even as other components of hematopathology practice, such as white blood cell differential interpretation, have been successfully automated.[6]

KEY POINTS

- In a mixed-methods quality improvement study, we developed and deployed a web-based clinical decision support tool, PROSER, to assist pathologists in peripheral blood smear (PBS) interpretation.
- PROSER interfaced with the hospital system's electronic health record to obtain and display relevant demographic, laboratory, and medication information and draft a PBS interpretation report.
- PROSER was used regularly by resident pathologists during the observation period and was overall well received.

KEY WORDS

Peripheral blood smear; Clinical decision support; Informatics; Laboratory medicine

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In 2011, Jaso et al[7] described a reimagined PBS interpretation workflow that incorporated a "web-based synoptic reporting system," which remains online over a decade later. This system contains a webpage with a series of annotated checkboxes, which, when selected, generate interpretive text derived from a library of prior PBS interpretations. As an example, a click on a checkbox for iron deficiency will generate an interpretive comment describing the likely PBS findings. In 2018, Kim et al[8] described an application that extended beyond simple documentation assistance to incorporate the numeric interpretation of laboratory data. This software interpreted the complete blood count and microscopic findings into a morphologic description and a corresponding follow-up recommendation. For example, the application assigned the term microcytic to a mean corpuscular volume of less than 80 fL, in accordance with International Council for Standardization in Haematology (ICSH) guidelines.[9] The follow-up recommendations were limited to six interpretations such as a request for a hemolysis workup (eg, lactate dehydrogenase, direct antiglobulin test, and haptoglobin) in response to documentation of schistocytes.

Both systems by Jaso et al[7] and Kim et al[8] employ structured reporting, which is the principle of using standardized, organized formats to clearly and accessibly convey information, often facilitated by templates, drop-down menus, or more complex computer programs. Structured reporting is used widely in both pathology[10,11] and radiology.[12-14] A 2016 systematic review found improvement in the reporting of clinically relevant information with the introduction of standardized, synoptic reporting.[11]

While the systems by Jaso et al[7] and Kim et al[8] offered improvements over existing workflows and promoted standardization of PBS reporting, neither system automated the integration of data beyond the CBC into an interpretation. Electronic health record (EHR) data, including historical ancillary laboratory values (eg, iron studies, flow cytometry), medications (eg, iron supplementation), and prior smear reviews, provide incredibly useful historical context to a PBS interpretation. In our institution, the time-consuming process of collecting these data, which is currently dispersed across the EHR, is the responsibility of the resident physician reviewing the PBS. Such clerical tasks have been associated with increased physician burnout and decreased job satisfaction.[15,16] Computer-based clinical decision support systems can offload such tasks by automating data collection and organizing information in a way that best supports clinical reasoning.[17] We therefore developed PROSER (PeRipheral blOod Smear EHR Reporter), a web-based software program that aids the pathologist in the documentation of PBS morphology, provides a historical context to the PBS with an organized display of EHR data, and composes an editable standardized, interpretive synthesis.

MATERIALS AND METHODS

Study Location/Setting

The study took place between August 1 and September 30, 2022, at a 200-bed academically affiliated Veterans Affairs hospital in the northeastern United States with an emergency department, medical and surgical floors, and an intensive care unit. Historical patient data originated from the hospital system's real-time EHR Database (Vista) and Corporate Data Warehouse (CDW). We followed the Standards for Quality Improvement Reporting Excellence 2.0 reporting guidelines.[18]

PROSER Design

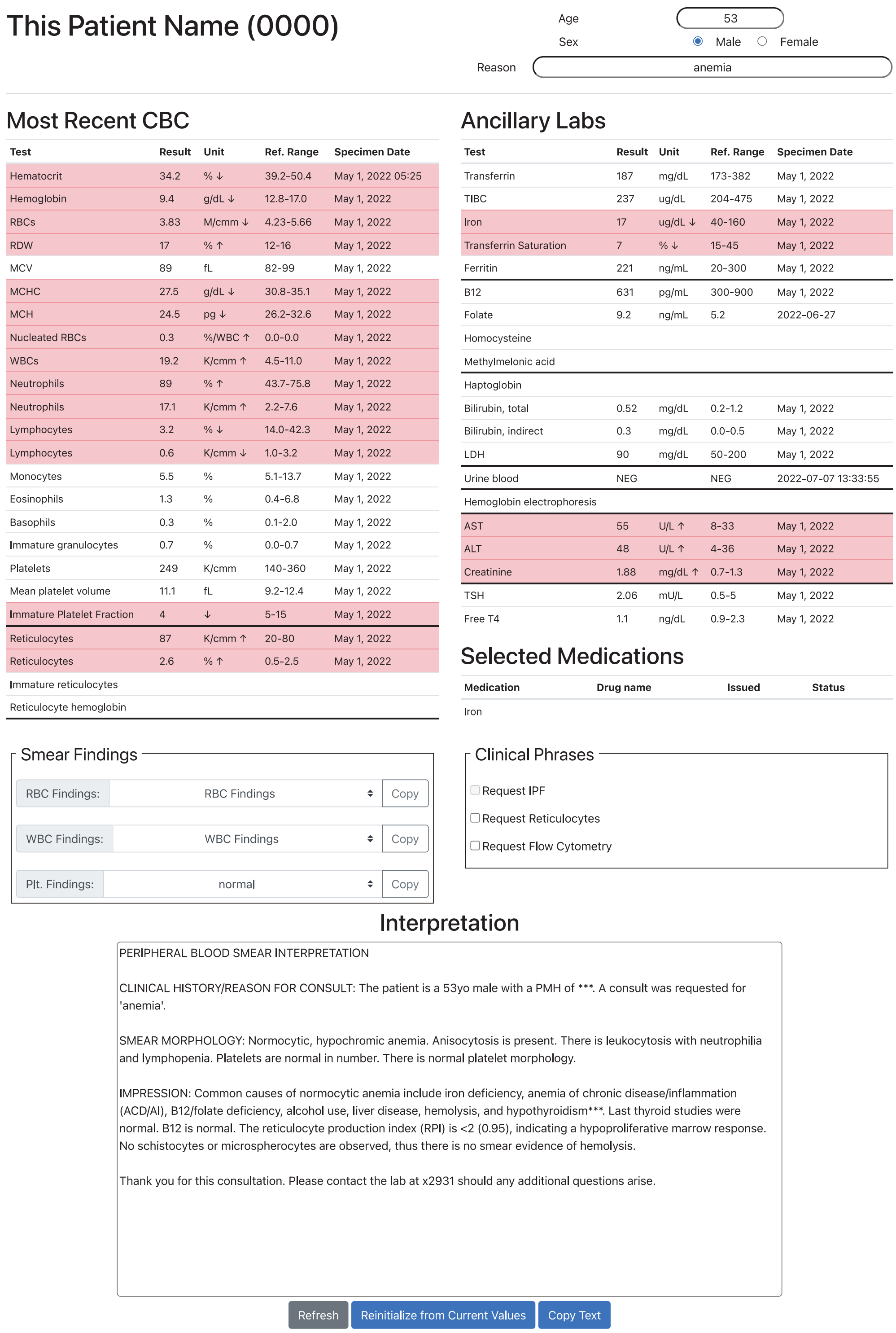
We created a web-based decision support tool that automated the clerical burden of data collection and provided simple rule-based logic to assist with PBS interpretation FIGURE 1. It assembled patients' EHR data from the Veterans Health Administration's (VHA's) data warehouse to display in a web-based graphical user interface. The EHR data were interpreted by a series of "if-then" rules, the output of which was displayed in a textbox. PROSER's features and functionality were selected based on review of existing smear interpretations and team consensus; the program was refined through an iterative Agile software development methodology incorporating feedback from resident and attending pathologist users.

A detailed flowsheet of the steps used to generate the impression, including a description of the potential phrases, is available in Supplemental Table 1 (all supplemental materials can be found at *American Journal of Clinical Pathology* online). As the pathologist viewed the peripheral blood smear, they input their morphologic findings through checkbox and radio button clicks to amend the interpretation. The pathologist reviewed the suggested interpretation, editing based on their clinical knowledge.

Data Flow

PROSER has 2 main software components, an Extract Transform Load (ETL) program and user-facing single-page web application. FIGURE 2 shows the data flow and roles of these components. First, PROSER queries the real-time EHR database to identify patients with a pending PBS. Second, PROSER obtains these patients' historical data from the CDW with a look-back period of 1 year. PROSER uses the historical data to create a patient-specific webpage. It stores the files for the single-page web application on a secure, shared drive with authentication controlled by Windows credentials. These pages are not accessible from the Internet but run stand-alone in standard web browsers. Pathologists with drive access open the web application, input additional information based on their microscopic review of the PBS, and edit PROSER's draft report, uploading a final report to the EHR. The ETL is written in C# using the Microsoft.NET framework (v4.7.2) and runs on a dedicated workstation. The single-page web application uses standard HTML components, Bootstrap for CSS styling (v4.4.1) and JavaScript (including JQuery v3.4.1), for interactive component scripting. The code for the entire PROSER project is available on GitHub, including an interactive demonstration of the PROSER user interface that allows the user to modify example laboratory values to view associated changes in PROSER output.

We obtained data from 2 sources: the EHR database, which is updated in real time but limited to our local site, and the CDW, which has comprehensive and reliable data from across Veterans Affairs (VA) sites but updates with a slight delay, a limitation shared



This Patient Name (0000)

Age 53
Sex Male Female
Reason anemia

Most Recent CBC

Test	Result	Unit	Ref. Range	Specimen Date
Hematocrit	34.2	% ↓	39.2-50.4	May 1, 2022 05:25
Hemoglobin	9.4	g/dL ↓	12.8-17.0	May 1, 2022
RBCs	3.83	M/cmm ↓	4.23-5.66	May 1, 2022
RDW	17	% ↑	12-16	May 1, 2022
MCV	89	fL	82-99	May 1, 2022
MCHC	27.5	g/dL ↓	30.8-35.1	May 1, 2022
MCH	24.5	pg ↓	26.2-32.6	May 1, 2022
Nucleated RBCs	0.3	%/WBC ↑	0.0-0.0	May 1, 2022
WBCs	19.2	K/cmm ↑	4.5-11.0	May 1, 2022
Neutrophils	89	% ↑	43.7-75.8	May 1, 2022
Neutrophils	17.1	K/cmm ↑	2.2-7.6	May 1, 2022
Lymphocytes	3.2	% ↓	14.0-42.3	May 1, 2022
Lymphocytes	0.6	K/cmm ↓	1.0-3.2	May 1, 2022
Monocytes	5.5	%	5.1-13.7	May 1, 2022
Eosinophils	1.3	%	0.4-6.8	May 1, 2022
Basophils	0.3	%	0.1-2.0	May 1, 2022
Immature granulocytes	0.7	%	0.0-0.7	May 1, 2022
Platelets	249	K/cmm	140-360	May 1, 2022
Mean platelet volume	11.1	fL	9.2-12.4	May 1, 2022
Immature Platelet Fraction	4	↓	5-15	May 1, 2022
Reticulocytes	87	K/cmm ↑	20-80	May 1, 2022
Reticulocytes	2.6	% ↑	0.5-2.5	May 1, 2022
Immature reticulocytes				
Reticulocyte hemoglobin				

Ancillary Labs

Test	Result	Unit	Ref. Range	Specimen Date
Transferrin	187	mg/dL	173-382	May 1, 2022
TIBC	237	ug/dL	204-475	May 1, 2022
Iron	17	ug/dL ↓	40-160	May 1, 2022
Transferrin Saturation	7	% ↓	15-45	May 1, 2022
Ferritin	221	ng/mL	20-300	May 1, 2022
B12	631	pg/mL	300-900	May 1, 2022
Folate	9.2	ng/mL	5.2	2022-06-27
Homocysteine				
Methylmelonic acid				
Haptoglobin				
Bilirubin, total	0.52	mg/dL	0.2-1.2	May 1, 2022
Bilirubin, indirect	0.3	mg/dL	0.0-0.5	May 1, 2022
LDH	90	mg/dL	50-200	May 1, 2022
Urine blood	NEG		NEG	2022-07-07 13:33:55
Hemoglobin electrophoresis				
AST	55	U/L ↑	8-33	May 1, 2022
ALT	48	U/L ↑	4-36	May 1, 2022
Creatinine	1.88	mg/dL ↑	0.7-1.3	May 1, 2022
TSH	2.06	mU/L	0.5-5	May 1, 2022
Free T4	1.1	ng/dL	0.9-2.3	May 1, 2022

Selected Medications

Medication	Drug name	Issued	Status
Iron			

Smear Findings

RBC Findings: RBC Findings Copy
WBC Findings: WBC Findings Copy
Plt. Findings: normal Copy

Clinical Phrases

Request IPF
Request Reticulocytes
Request Flow Cytometry

Interpretation

PERIPHERAL BLOOD SMEAR INTERPRETATION

CLINICAL HISTORY/REASON FOR CONSULT: The patient is a 53yo male with a PMH of ***. A consult was requested for 'anemia'.

SMEAR MORPHOLOGY: Normocytic, hypochromic anemia. Anisocytosis is present. There is leukocytosis with neutrophilia and lymphopenia. Platelets are normal in number. There is normal platelet morphology.

IMPRESSION: Common causes of normocytic anemia include iron deficiency, anemia of chronic disease/inflammation (ACD/AI), B12/folate deficiency, alcohol use, liver disease, hemolysis, and hypothyroidism***. Last thyroid studies were normal. B12 is normal. The reticulocyte production index (RPI) is <2 (0.95), indicating a hypoproliferative marrow response. No schistocytes or microspherocytes are observed, thus there is no smear evidence of hemolysis.

Thank you for this consultation. Please contact the lab at x2931 should any additional questions arise.

Refresh | Reinitialize from Current Values | Copy Text

FIGURE 1 Example PROSER website displaying relevant patient data, options for physician-entered smear findings and clinical phrases, and draft smear interpretation. Values outside of the laboratory reference range are highlighted, with arrows indicating the direction of the abnormality.

by data warehouses in general. Unlike many EHR data warehouses, the CDW does not update nightly but instead does so continuously at irregular intervals, as server time allows, with a typical transport time of less than 24 hours. The time required from consult placement to peripheral smear availability, including the patient's blood draw and slide preparation, averaged 1 day in our laboratory. Therefore, the CDW typically had up-to-date data at the time of peripheral smear interpretation and was selected as the data source for laboratory and other historical data due to its reliability and inclusion of data from all VA sites. To ensure that a website had been created for

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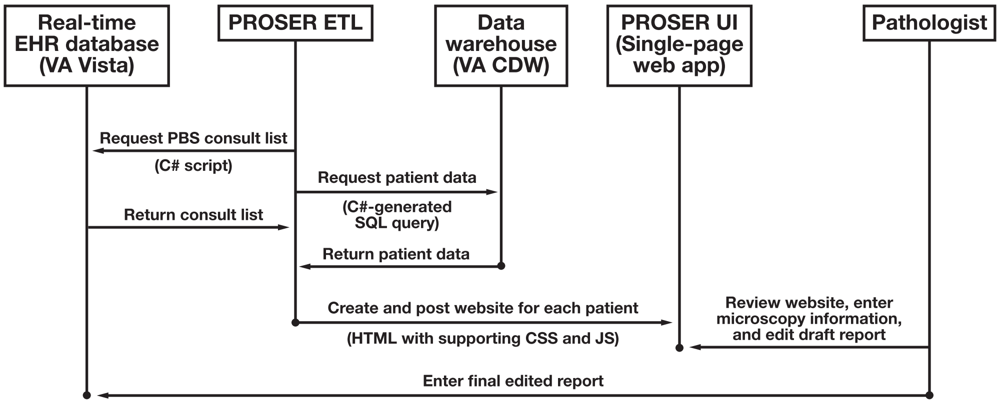

FIGURE 2 PROSER interactions with software resources and the pathologist. CDW, Clinical Data Warehouse; CSS, Cascading Style Sheets; EHR, electronic health record; ETL, extract, transform, and load; JS, JavaScript; PBS, peripheral blood smear; SQL, Structured Query Language; VA, Veterans Association.

each patient with an active PBS consult at our clinical site, PROSER was automated to pull the consult list directly from the local EHR database.

Data warehouse elements included the reason for the consult request, the patient's demographics, laboratory results, and medications. We selected the data elements, viewable in FIGURE 1, based on the most common data referenced by our local pathologists during PBS interpretation. We sought to provide the most relevant information rather than an exhaustive list, with the goal of minimizing clutter. The pathologist reviewing the PBS could consult the EHR for additional data at their discretion.

Clinical Workflow

Workflow in our institution begins with resident physician interpretations of the PBS. Therefore, resident physicians primarily engaged with PROSER, while attending physicians rarely used PROSER directly. The resident performed microscopic review of the PBS, input additional information into the website based on their findings, and reviewed and edited the website's suggested consult report. Attending physicians then reviewed and, if necessary, modified the resident physician's draft prior to entering the final report into the EHR. Data were obtained and analyzed as part of an operational quality improvement project rather than a systematic investigation designed to develop or contribute to generalizable knowledge; this project was therefore exempt from the requirements of the Common Rule and did not require institutional review board review. Drafting and submission of this article complied with applicable VHA policies (VHA Program Guide 1200.21, VHA Operations Activities That May Constitute Research).[19]

Assessment of Impact

We conducted a user survey to assess the impact of PROSER on PBS interpretation, based on the survey by Jaso et al[7] for a synoptic reporting system for PBS interpretation. We administered the survey to residents and attending physicians at the conclusion of their PBS rotation. The survey and responses are shown in TABLE 1; in addition to these survey questions, respondents were given the option of adding additional comments in a free text response.

RESULTS

Interpretation of Data

In constructing the automated smear impression, PROSER used a collection of "if-then" rules to translate laboratory findings into plain language, calculate relevant indices, insert relevant prewritten lists of differential diagnoses, and recommend additional workup that might help clarify a diagnosis. PROSER displayed 46 laboratory values with corresponding reference ranges and abnormal flags, allowed for entry of 14 microscopy findings, and computed 2 calculations based on laboratory values; this information informed the automated composition of a draft PBS report, using a library of 92 prewritten phrases. Examples of PROSER's logic can be found in TABLE 2 and FIGURE 1. Additional impressions and recommendations could be manually entered by the pathologist.

In the absence of a gold standard for peripheral blood smear consultation responses, the structure and verbiage of the interpretations were modeled on existing local practice, with rule logic based on expert opinion, hematology literature, laboratory reference ranges, and ICSH-suggested terminology.[9] Interpretations comprised a header, an automated description of smear morphology, an automated smear impression, and a standardized footer; an example can be seen in FIGURE 1.

While PROSER did not have access to actual PBS images, it inferred likely PBS findings from laboratory data and used additional microscopy information entered by the pathologist, incorporating both automated and physician-entered data in its suggested report. It was the responsibility of the resident and attending pathologist to perform both microscopic evaluation of the PBS slide and chart review, editing PROSER's suggested text as needed to incorporate their findings, correct any erroneous interpretations, and ensure that any suggested interpretation (eg, a list of differential diagnoses) was relevant to the specific patient's clinical history and context. PROSER exists for download and modification on our public GitHub repository at https://ajloza.github.io/PeripheralBloodSmear/.[20] A demonstration version of PROSER is available on the GitHub repository.

This version is modified from the production version to allow users to modify laboratory values and view the resulting changes to the PROSER-generated interpretation. A detailed use guide is available on the main GitHub repository page.

Use

During the 2-month observation period, 6 resident physicians overseen by 4 attending physicians entered 189 PBS interpretations on a veteran population of 92% (173/189) males with a median age of 73 (interquartile range, 64-78) years. Several minor updates were made to PROSER during the study period, primarily additions or refinements of interpretive text and logic, driven by resident physician feedback.

Survey

Five (83%) of 6 residents and 2 (67%) of 3 surveyed attending physicians completed the survey. One attending physician reported never interacting with PROSER and therefore abstained. Another attending physician did not receive a survey due to abstention as the paper's senior author.

Residents reported using PROSER always (60%) or often (40%). All residents reported that PROSER positively affected PBS interpretation quality (80% "very positively," 20% "somewhat positively"), PBS interpretation standardization (60% "very positively," 40% "somewhat positively"), education (80% "very positively," 20% "somewhat positively"), and enjoyment of work (40% "very positively," 60% "somewhat positively"). They reported that

TABLE 1 Survey Questions and Results

Question	Response	Resident physician (n = 5), No. (%)	Attending physician (n = 2), No. (%)
How often did you use PROSER while reading peripheral blood smears during the study period?	Always	3 (60)	
	Often	2 (40)	1 (50)
	Sometimes		
	Rarely		1 (50)
	Never		
How did PROSER affect the standardization of PBS interpretation?	Very negatively		
	Somewhat negatively		
	Neutrally/no effect		
	Somewhat positively	2 (40)	2 (100)
	Very positively	3 (60)	
How did PROSER affect the quality of PBS interpretation?	Very negatively		
	Somewhat negatively		
	Neutrally/no effect		1 (50)
	Somewhat positively	1 (20)	1 (50)
	Very positively	4 (80)	
How did PROSER affect the frequency of typographical, spelling, and grammatical errors?	Very negatively		
	Somewhat negatively		
	Neutrally/no effect	2 (40)	
	Somewhat positively		2 (100)
	Very positively	3 (60)	
How did PROSER affect the extent to which you enjoyed your work?	Very negatively		
	Somewhat negatively		
	Neutrally/no effect		2 (100)
	Somewhat positively	3 (60)	
	Very positively	2 (40)	
How did PROSER affect your education during this rotation?	Very negatively		
	Somewhat negatively		1 (50)
	Neutrally/no effect		1 (50)
	Somewhat positively	1 (20)	
	Very positively	4 (80)	
How often did you experience any safety concerns while using PROSER?	Always		
	Often		
	Sometimes		
	Rarely	1 (20)	1 (50)
	Never	4 (80)	1 (50)
How often did you find PROSER had missing data?	Always		
	Often		1 (50)
	Sometimes	3 (60)	
	Rarely	2 (40)	
	Never		1 (50)

PBS, peripheral blood smear.

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TABLE 2 Examples of PROSER's Functionality

Type of functionality	Example circumstance/input	Output
Plain-language cell morphology descriptions		
Cell line abnormalities	•Hemoglobin is low[a] •White blood cell count is low •Platelets are within reference range	"Bicytopenia is present with anemia and leukopenia."
Detailed erythrocyte abnormalities	•Hemoglobin is low •MCV is low •MCHC is low	"Microcytic, hypochromic, anemia."
Ancillary laboratory interpretations		
Immature platelet fraction	•Platelets are low •Immature platelet fracture is low or within normal limits	"The immature platelet fraction is not elevated, even though the patient has thrombocytopenia. This suggests a deficit in platelet production."
Iron studies in microcytic anemia	•Hemoglobin is low •MCV is low •Ferritin is low •Transferrin is high	"The low ferritin and high transferrin suggest iron deficiency is more likely than ACD/AI."[21]
Incorporation of information input via check boxes		
Interpretation of clinician-entered microscopy findings	•Hemoglobin is low •MCV is low •Ferritin is low	"No schistocytes or microspherocytes are observed, thus there is no smear evidence of hemolysis."
Request for additional studies	•Clinician selects box for flow cytometry request	"Recommend obtaining flow cytometry."
Special calculations		
Reticulocyte production index	•Hematocrit is low •Reticulocyte percentage is reported •Reticulocyte production index is <2	"The reticulocyte production index (RPI) is <2, indicating a hypoproliferative marrow response."
Mentzer index[22] in microcytic anemia	•Hemoglobin is low •MCV is low •RBC count is normal	"The Mentzer index is <13, favoring thalassemia trait over iron deficiency."[22]
Differential diagnoses		
Microcytic anemia	•Hemoglobin is low •MCV is low	"Common causes of microcytic anemia include iron deficiency, thalassemia, and anemia of chronic disease/inflammation (ACD/AI)***."[b]
Thrombocytosis	•Platelet count is elevated	"Thrombocytosis can result from a reactive process or myeloproliferative neoplasm (MPN). Common reactive causes include inflammation (eg, infection, rheumatological condition, malignancy), iron deficiency, and post-splenectomy state. MPNs (eg, essential thrombocythemia, polycythemia vera, primary myelofibrosis) most commonly have mutations in JAK2, CALR, or MPL***."[b]

ACD, anemia of chronic disease; AI, anemia of inflammation; MCHC, mean corpuscular hemoglobin concentration; MCV, mean corpuscular volume.
[a] The terms high and low refer to laboratory values above and below the laboratory reference range, respectively.
[b] In the PROSER interface, a series of 3 asterisks is intended to prompt the clinician to complete a phrase or consider its relevance to the case at hand. To facilitate editing, the cursor can be brought to an instance of "***" through the keyboard shortcut F2.

PROSER affected the frequency of typographical, spelling, and grammatical errors "very positively" (60%) or neutrally (40%). Missing data were reported "sometimes" (60%) or "rarely" (40%), and safety concerns were reported "never" (80%) or "rarely" (20%).

One attending physician reported using PROSER "often" and claimed that it "somewhat positively" affected PBS standardization, quality, and the frequency of typographical errors, with "neutral" effect on enjoyment of work and resident education. This attending reported "never" experiencing missing data or having safety concerns. The other attending reported "rarely" using PROSER and indicated that it "somewhat positively" affected PBS standardization and the frequency of typographical errors, with "neutral" effect on PBS quality and enjoyment of work and "somewhat negative" effect on resident education. This attending reported "often" experiencing missing data and "rarely" having safety concerns. Survey results are shown in **TABLE 1**.

Free-text responses were generally positive. One resident described the program as "overall a great starting point to look at important data needed for interpretation," and another described it as "overall … a great supplement for smear interpretation." The 2 attending physicians highlighted both strengths and challenges of the program. Regarding resident education, 1 attending reported that PROSER "helped [residents] think systematically about the cases," and another stated that "some residents used the template thoughtlessly; others used it effectively." One attending stated, "I least like the somewhat monotonous and pedantical list of differentials." The other noted that "the best feature is the automated display of relevant lab values."

DISCUSSION

PROSER represents a formative step toward improved computer-assisted data integration and structured reporting in pathology report composition, building on prior work in PBS interpretation. PROSER's success at gathering and displaying data and drafting interpretations is evidenced by its positive reviews from resident physicians, the primary end users who would otherwise perform these tasks. The thorough use of information from the data warehouse to contextualize a patient's PBS findings differentiates PROSER from prior work.

The positive sentiment reported by its users may reflect the benefits of delegating clerical and algorithmic tasks of PBS interpretation to PROSER, allowing the pathologist to focus on critical thinking while simultaneously improving the standardization of the report. The difference in reported frequency of use of the program between resident and attending physicians likely reflects our local workflow, in which resident physicians typically draft the initial PBS report, while attendings review the resident draft rather than beginning de novo.

Limitations

The logic used to write PROSER's interpretations may not apply to all population or settings. In particular, aspects of PROSER's logic may not perform equally well in women and children as in our majority male, all-adult population. Although this represents a limitation, few steps in the interpretation logic are specific to sex or age. Most logic compares the patient's test result with the corresponding laboratory reference range; since PROSER consumes both the result and reference range from the source data, it would take and sex- or age-specific reference ranges into account. The interpretations provided by PROSER may also represent our local dialect and practice patterns. To allow others to optimize PROSER for various practice settings and populations, we created extensible code with clinical phrases isolated to a single text file that can be edited as needed.[20]

While our user survey was anonymous, it is possible that, due to the small sample size and professional relationships between the survey participants and PROSER developers, survey results were positively skewed by so-called response bias or that participants with negative views were less likely to respond (an example of "nonresponse bias").[23]

Future Directions

We anticipate 2 enhancements to future iterations of PROSER that would make the program more effective. First, future designs could feed PROSER data directly from the EHR or through an HL7 message router, instead of or in addition to our current data source, the CDW; the incorporation of additional real-time data could address issues related to missing data and associated safety concerns. Second, PROSER's current set of rules could extend to additional concepts, disease processes, and guidelines. Future applications of PROSER or similar software at other sites must consider and address site-specific data access protocols and concerns to ensure patient data security.

Looking ahead, multiple lines of research have the potential to dramatically change PBS interpretation, including computer-assisted image interpretation, computer-assisted integration of a patient's medical history or data trends, and dashboards for pathologists to interactively engage with these tools. Automated capture of blood cell images already exists.[24] Incorporation of these images into PROSER would increase its functionality, perhaps greatly reducing the effort required for manual smear review.

Future work should evaluate PROSER's clinical efficacy, including both clinician-level and patient-level outcomes. Clinician-level outcomes could include time spent (or saved) on information gathering and PBS interpretation, the proportion of final PBS interpretation text that originated from PROSER, and the subjective value of the PBS interpretation to the ordering physician. Patient-level outcomes could include rates of downstream testing and diagnosis and clinical outcomes.

CONCLUSIONS

In this quality improvement study, we successfully developed and deployed a web-based clinical decision support tool, PROSER, to assist pathologists with PBS interpretation. A small user survey revealed that PROSER was well received by resident physicians. Future work should further refine this intervention and quantitatively evaluate its effects on patient-centered clinical outcomes and resident training.

Conflict of interest disclosure: The authors have nothing to disclose.

REFERENCES

1. ACGME. ACGME Program Requirements for Graduate Medical Education in Anatomic Pathology and Clinical Pathology. 2022. Accessed July 18, 2022. https://www.acgme.org/globalassets/pfassets/programrequirements/300_pathology_2020.pdf
2. Chaparro CM, Suchdev PS. Anemia epidemiology, pathophysiology, and etiology in low- and middle-income countries. *Ann N Y Acad Sci.* 2019;1450:15-31. https://doi.org/10.1111/nyas.14092
3. Gnanaraj J, Parnes A, Francis CW, et al. Approach to pancytopenia: diagnostic algorithm for clinical hematologists. *Blood Rev.* 2018;32:361-367. https://doi.org/10.1016/j.blre.2018.03.001
4. Adewoyin AS, Nwogoh B. Peripheral blood film—a review. *Ann Ib Postgrad Med.* 2014;12:71-79.
5. Beckman AK, Ng VL, Jaye DL, et al. Clinician-ordered peripheral blood smears have low reimbursement and variable clinical value: a three-institution study, with suggestions for operational efficiency. *Diagn Pathol.* 2020;15:112. https://doi.org/10.1186/s13000-020-01033-8
6. Barnes PW, McFadden SL, Machin SJ, et al; International Consensus Group for Hematology. The International Consensus Group for Hematology Review: suggested criteria for action following automated CBC and WBC differential analysis. *Lab Hematol.* 2005;11:83-90. https://doi.org/10.1532/LH96.05019

7. Jaso J, Nguyen A, Nguyen AN. A synoptic reporting system for peripheral blood smear interpretation. *Am J Clin Pathol*. 2011;135:358-364. https://doi.org/10.1309/AJCPPGSA9D0HYNAH
8. Kim YG, Kwon JA, Moon Y, et al. An automated draft report generator for peripheral blood smear examinations based on complete blood count parameters. *Ann Lab Med*. 2018;38:512-517. https://doi.org/10.3343/alm.2018.38.6.512
9. Palmer L, Briggs C, McFadden S, et al. ICSH recommendations for the standardization of nomenclature and grading of peripheral blood cell morphological features. *Int J Lab Hematol*. 2015;37:287-303. https://doi.org/10.1111/ijlh.12327
10. Ellis DW, Srigley J. Does standardised structured reporting contribute to quality in diagnostic pathology? The importance of evidence-based datasets. *Virchows Arch*. 2016;468:51-59. https://doi.org/10.1007/s00428-015-1834-4
11. Sluijter CE, van Lonkhuijzen LR, van Slooten HJ, et al. The effects of implementing synoptic pathology reporting in cancer diagnosis: a systematic review. *Virchows Arch*. 2016;468:639-649. https://doi.org/10.1007/s00428-016-1935-8
12. Powell DK, Silberzweig JE. State of structured reporting in radiology, a survey. *Acad Radiol*. 2015;22:226-233. https://doi.org/10.1016/j.acra.2014.08.014
13. Nobel JM, Kok EM, Robben SGF. Redefining the structure of structured reporting in radiology. *Insights Imaging*. 2020;11:10. https://doi.org/10.1186/s13244-019-0831-6
14. Lin E, Powell DK, Kagetsu NJ. Efficacy of a checklist-style structured radiology reporting template in reducing resident misses on cervical spine computed tomography examinations. *J Digit Imaging*. 2014;27:588-593. https://doi.org/10.1007/s10278-014-9703-2
15. West CP, Dyrbye LN, Shanafelt TD. Physician burnout: contributors, consequences and solutions. *J Intern Med*. 2018;283:516-529. https://doi.org/10.1111/joim.12752
16. Peccoralo LA, Kaplan CA, Pietrzak RH, et al. The impact of time spent on the electronic health record after work and of clerical work on burnout among clinical faculty. *J Am Med Inform Assoc*. 2021;28:938-947. https://doi.org/10.1093/jamia/ocaa349
17. Musen MA, Middleton B, Greenes RA. Clinical decision-support systems. In: Shortliffe EH, Cimino JJ, eds. *Biomedical Informatics: Computer Applications in Health Care and Biomedicine*. Springer International Publishing; 2021:795-840.
18. Ogrinc G, Davies L, Goodman D, et al. SQUIRE 2.0 (Standards for QUality Improvement Reporting Excellence): revised publication guidelines from a detailed consensus process. *BMJ Qual Saf*. 2016;25:986-992. https://doi.org/10.1136/bmjqs-2015-004411
19. Department of Veterans Affairs Office of Research & Development. Program Guide: 1200.21. VHA Operations Activities That May Constitute Research. Accessed January 9, 2019. https://www.research.va.gov/resources/policies/ProgramGuide-1200-21-VHA-Operations-Activities.pdf
20. PROSER. Version 1. 2022. Accessed February 23, 2023. https://github.com/hauserrg/PeripheralBloodSmear
21. Weiss G, Goodnough LT. Anemia of chronic disease. *N Engl J Med*. 2005;352:1011-1023. https://doi.org/10.1056/NEJMra041809
22. Mentzer WC Jr. Differentiation of iron deficiency from thalassaemia trait. *Lancet*. 1973;1:882. https://doi.org/10.1016/s0140-6736(73)91446-3
23. Grimm P. Social desirability bias. In: Sheth JN, Malhotra NK, eds. *Wiley International Encyclopedia of Marketing*. John Wiley & Sons Ltd; 2010.
24. Mohammed EA, Mohamed MM, Far BH, et al. Peripheral blood smear image analysis: a comprehensive review. *J Pathol Inform*. 2014;5:9. https://doi.org/10.4103/2153-3539.129442